



# **Federal Republic of Ethiopia**

## **Ministry of Education**

### **B.Sc. in Software Engineering Program National Exit Examination**

#### **Identified Competency Focus Areas and Core Courses**

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## **1. Introduction**

To fulfill demands of software engineering professionals in various local and international industries, it has become necessary to produce graduates equipped with the required knowledge, skills, and attitude. This has necessitated the need to evaluate the knowledge, attitude, and skill set of the software engineering graduates using the identified competencies and focus areas of the software engineering curriculum. As software engineering program is one of the most evolving and demanded in this era, the stakeholders who are taking part in this program should mandate themselves to work towards ensuring the knowledge, attitude, and skill set of graduates. One of key benefits of evaluating and certifying the graduates based on the competency requirements of software engineering, to win the trust of the industries to consume and employ the graduates, and to introduce a sense of hard work in the mindset of the graduates early in their academic years. This will enable them to explore and dig further for the relevant skill sets of software engineering program that are needed by the market. The intent of this draft document for software engineering is to provide a framework within which to conceptualize the competency areas that should be covered and addressed by the exit exam to be prepared in the future. Overall, this guideline of software engineering exit examination for Ethiopian higher institutions has been prepared relying on the new existing curricula of Addis Ababa Science and Technology University (AASTU) and Arba Minch University (AMU), ultimately aiming to maintain the program and graduate quality through evaluating graduates' skill sets to increase trust of employers' from local and international industries . Overall, the software engineering exit exams for Ethiopian higher institutions have been prepared with the following objectives in mind:

### **a. Objectives**

The national exit exam for software engineering program shall have the following objectives

- Producing skilled and competent manpower to national and international market
- Assessing students' educational achievement in major areas of software engineering

- Ensuring whether the graduation profile of software engineering curriculum have achieved at least common standards of knowledge and practical skills
- Improving public trust and confidence in software engineering activities of professionals
- Facilitating the efforts of students to revise the core learning outcomes of the courses covered by the exit examination
- Ensuring all graduates from HEIs satisfy the requirements of the labor market and employability through the national wide implementation of competency-based exit exam
- Creating competitive spirit among software development/application sectors in Ethiopia with encouraging them to give due attention to the national standards

**b. Significance:**

It is important to set competency areas of the subject matter (program) in order to measure the how much graduates are acquired with skills, knowledge and attitudes. The following shows us the significance or setting competencies and identifying core courses of the program;

- To set competencies that helps to assess the basic skills, knowledge and attitude of graduating students;
- To systematically identify the core courses which will be included the exit exam;

**2. Expected Profile of Graduates**

This exit exam is expected to measure the software engineering graduates knowledge, skill and attitude so that they:

- Use current techniques, skills, and tools necessary for software development, testing, and maintenance.
- Acquire the knowledge and innovative skills to plan, set up and run software related businesses.
- Understand best practices and standards of software engineering program and their application.

- Be familiar with the required software project management skill that enable them to manage complex software projects
- Understand professional, ethical, legal, security and social issues and responsibilities.
- Have ability to apply knowledge of computing and mathematics to resolve on-hand and upcoming problems.
- Have recognition of the need for, and an ability to engage in, continuing professional development and the knowledge and skills to act as research assistants or lecturers in higher education institutions

### **3. Competencies and Learning Outcomes**

Software engineering graduates are expected to demonstrate the following knowledge, skill and attitudes:

#### **i. Knowledge**

- Conduct investigation of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions
- Adopt, innovate, select, and apply appropriate state of the art software design and development tools, methodologies, resources, and emerging technologies including simulation, prediction and modeling for complex business process with understanding of their potential capabilities, limitations and associated risks.
- Identify, formulate, research literature, and analyze complex problems, reaching substantiated (justifiable, validated) conclusions using software development principles, methodologies, and tools.
- Apply logical reasoning and informed decision to assess societal, health, safety, privacy, security, legal and cultural issues, and the consequent responsibilities relevant to software engineering application
- Understand the impact of software applications in societal and environmental contexts and apply the principle of green computing for sustainable development.

## **ii. Skill**

- Design integrated and efficient software solutions, component, or processes to address complex problems and implement them to meet the specified needs with appropriate consideration for public health and safety, cultural, societal (local and international) and environmental considerations.
- Communicate effectively on complex business and system activities with the software engineering community and with a society at large; such as being able to comprehend and write effective reports, articles and design documentations using various tools, make effective presentations and give and receive clear instructions.
- Demonstrate knowledge and skill on software project management principles and apply these to manage software development processes and its resource requirements, costs in multidisciplinary projects
- Apply knowledge of mathematics, and software engineering specialization to address complex and large-scale problems.

## **iii. Attitude**

- Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.
- Function effectively as an independent software engineer, as a member or as a leader in a diverse team and multidisciplinary settings.
- Recognize the need for and have the preparation as well as ability to engage in an independent and lifelong learning in the information and knowledge economy.

## **4. Categorizing Courses into Themes(4-6) and corresponding Courses (10-15)**

To help demarcate the focus areas and competency boundaries that are covered by the identified courses, the courses have been categorized under six themes. Accordingly, Table 1 maps courses to the respective themes identified.

Table 1: Courses categorization based on themes

| S/<br>N            | Themes                                                         | Courses to be included in the Exit Exam                                | Cred.hrs  |
|--------------------|----------------------------------------------------------------|------------------------------------------------------------------------|-----------|
| 1                  | Problem analysis and programming                               | 1. *Fundamental of Programing                                          | 8         |
|                    |                                                                | 2. Fundamental of Data structure and Algorithm                         | 4         |
|                    |                                                                | 3. Object Oriented Programming                                         | 3         |
|                    |                                                                | 4. Internet Programming (or Web Programming)                           | 6         |
|                    |                                                                | 5. Mobile Application Development                                      | 3         |
|                    |                                                                | 6. Fundamentals of Database Systems                                    | 4         |
|                    |                                                                | 7. Operating Systems                                                   | 4         |
| 2                  | Software requirement, Design, and Architecture                 | 8. Fundamentals of Software Engineering                                | 4         |
|                    |                                                                | 9. Software Requirement Engineering                                    | 3         |
|                    |                                                                | 10. Software Architecture and Design                                   | 3         |
| 3                  | Software project management, maintenance and quality assurance | 11. Software Project Management                                        | 3         |
|                    |                                                                | 12. Software Testing and Quality Assurance                             | 3         |
|                    |                                                                | 13. Software Evolution and Maintenance                                 | 3         |
| 4                  | Networking and Software Security                               | 14. *Fundamentals of Networking, and Software and information security | 7         |
| 5                  | AI and related Areas                                           | 15. *Fundamentals of Artificial Intelligence and Machine Learning      | 6         |
| Total Credit Hours |                                                                |                                                                        | <b>64</b> |

\* Fundamental of Programing I(4crhr) and Fundamental of Programing II(4crhrs)

\* Fundamentals of Networking (4crhr), and Software and information Security(3crhr)

\*Fundamentals of Artificial Intelligence (3chrs) and Machine Learning (3crhr)

## 5. Conclusion

It is apparent that assessing graduates based on the curriculum's core competencies and industry demands will be crucial to increasing employers' confidence in their abilities. To realize this, it has become relevant to have a guideline that could be used as a reference to prepare a

comprehensive exit examination that could assess the relevant focus areas of the software engineering program. To this end, this document was prepared with the intent to highlight the core competency areas of the software engineering curriculum. Besides, the preparation of this document is also taking into account the anticipating market demand of the software engineering professionals for both the local and international industries. It aims to evaluate the graduates' knowledge, skills, and attitudes towards that end. Finally, it is assumed that the outcome of the software engineering exit examination could be used as a feedback loop for policymakers and to those who are engaged in curriculum development by modifying or incorporating new competencies into the curricula. It could also provide feedback so as to take other remedial actions that might improve the software engineering graduates skill set. As a recommendation, we suggested that it is good practice to communicate the exam content areas and scopes for each course to the students ahead of the exit exam. This would guide students where they need to focus much time on some courses with extended and vast contents. Also, as an exceptional recommendation, we suggest the relevant body entertain some universities for exceptional cases, as shown in the remarks section. In order to successfully assess the minimum requirements of the graduate profile in software engineering program, the basic core courses are selected, and those courses need to be considered for preparing the exit exam and notifying candidates prior to the exam delivery.

## 6. Remark

This section presents courses which brought differences in the course of the curricula of software engineering of the two universities (i.e. AASTU and AMU). The table below comprises the course, key difference in two curricula and suggested actions to be taken.

| SNo | Course                    | Differences in two Curricula                   | Suggestions                                                             |
|-----|---------------------------|------------------------------------------------|-------------------------------------------------------------------------|
| 1   | Software component design | Is a core course in AASTU, but elective in AMU | It can assessed in SW design and Architecture                           |
| 2   | Software metrics          | Is a core course in AMU, but absent in AASTU   | It can assessed in software project management, testing and maintenance |

|   |                         |                                                                                                  |                                                                 |
|---|-------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 3 | Software configuration  | Is a core course in AASTU, but is as a topic in software evolution and maintenance course of AMU | It can assessed under software evolution and maintenance course |
| 4 | AI and Machine learning | Are elective courses in the old curr.                                                            | Are as core courses in New curr.                                |

The remarks mentioned above shall be notified to those candidates who are taking software engineering exit exam. Besides, the department shall take necessary preparation for providing tutorial services for those in needs.